

Module 2

Basics of Signals and Transmission

Syllabus

- Analog Vs. Digital signals
 - Bandwidth and Frequency Spectrum
 - Transmission media: Wired (Copper, Fiber-optic) and Wireless
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2.1 Introduction to Signals

A signal is the fundamental component in any communication system. A signal is any physical quantity that carries information. Signals communicate information through variations in some measurable property such as voltage, current, light intensity, amplitude, frequency, or phase of EM waves.

In telecommunication systems, signals are usually electrical, optical signals or electromagnetic waveforms that travel through copper wires, optical fibers, or wireless channels to convey information.

The information can originate from a source as sound, light, temperature, or mechanical motion.

Significance: Signals are the fundamental carriers of information. They transform the original information into a form (electrical / optical / electromagnetic) that can be processed, amplified, and reliably transmitted across a channel.

Transformation: The original physical information must be converted into a measurable electrical quantity (voltage or current) by an Input Transducer (e.g., a microphone) before it can enter the telecommunication system for processing.

A well-designed communication system ensures that signals:

- Represent information accurately
- Travel with minimum loss or distortion
- Resist noise
- Reach the destination within acceptable quality levels

Types of Signals

Signals are fundamentally classified into two types based on how they vary over time.

1. Analog Signals
2. Digital Signals

2.2 Analog Signals

Definition: An analog signal is a continuous-time signal whose amplitude varies smoothly over time. The signal's electrical characteristics (such as voltage or current) are directly proportional to the physical information they represent, such as sound pressure or light intensity.

Nature: Its nature is continuous in both time and amplitude.

Waveform: While complex analog signals like voice are irregular, they are often mathematically represented by the basic building block, the sine wave.

Characteristics:

- *Nature:* Continuous in time and amplitude
- *Infinite Values:* The signal can take an infinite number of possible amplitude values within a given range.
- *Natural Representation:* They are ideal for representing natural information directly, such as sound, temperature, and light.
- *High Susceptibility to Noise:* They are highly susceptible to noise and interference during transmission.
- *Permanent Degradation:* Any noise added to an analog signal is amplified along with the signal by repeaters, permanently degrading the quality.
- *Bandwidth Inefficiency:* They consume bandwidth less efficiently than digital signals.

Advantages

- *Natural representation:* Analog signals naturally mimic real-world physical quantities such as sound, light, and temperature, allowing them to represent these phenomena accurately without complex conversion.
- *Simple generation and processing:* Analog signals can be generated and processed easily using basic electronic components like resistors, capacitors, and transistors, making analog circuits straightforward and cost-effective.

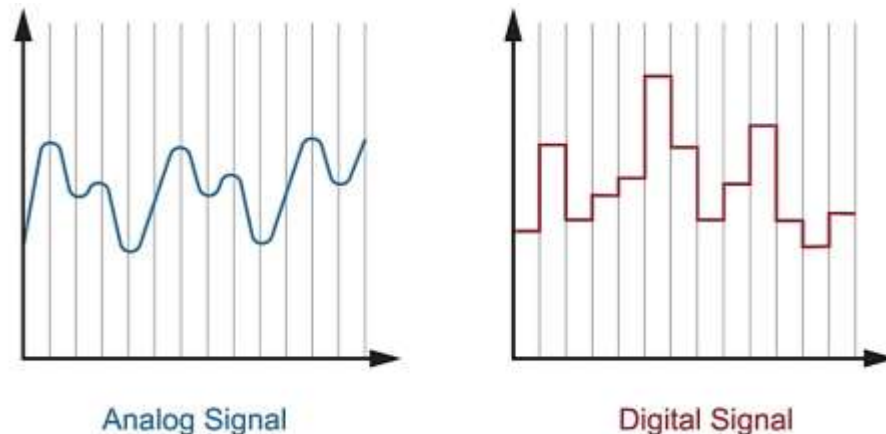
- *Continuous and smooth waveform:* Because analog signals vary smoothly over time without abrupt transitions, they provide a continuous waveform that preserves fine details of the original information.

Limitations:

- *High Susceptibility to Noise:* Analog signals are highly susceptible to noise and interference (e.g., static, crosstalk) during transmission.
- *Permanent Degradation:* Any noise added to an analog signal is amplified along with the signal by repeaters, leading to permanent degradation. The signal quality degrades visibly with distance.
- *Bandwidth Inefficiency:* They consume bandwidth less efficiently than digital signals when transmitting high-volume, complex data.
- *Poor long-distance reliability:* As analog signals travel over long distances, they continuously lose strength and accumulate noise, making it difficult to maintain high-quality transmission without significant degradation.
- *Security and encryption are difficult:* Because analog waveforms are continuous and cannot be encoded into discrete values, storing, processing, and encrypting them becomes challenging and less efficient compared to digital signals.
- *Less bandwidth efficient:* Analog transmission techniques typically require more bandwidth to carry high-quality signals, making them less efficient in modern communication systems where bandwidth is valuable.

Example: Human voice, traditional radio waves (AM/FM), voltage produced by a microphone.

Applications: FM/AM radio, analog TV.



2.3 Digital Signals

Definition: A digital signal is a discrete time signal, represented by a sequence of discrete pulses. It is most commonly a binary signal, meaning it consists of only two voltage levels: High (1) and Low (0).

Nature: Discrete (stepped) in both time and amplitude. It changes instantly from one level to the next.

Waveform: It is often represented as a square wave or a sequence of rectangular pulses, changing instantly from one level to the next.

Characteristics:

- *Finite Values:* The signal can only take on a finite set of values (e.g., 0 or 1).
- *Binary Representation:* Information is represented as a sequence of binary bits (0s and 1s).
- *High Noise Immunity:* Digital signals are highly resistant to noise. Repeaters can perfectly regenerate the original pulse shape and timing, eliminating accumulated noise over long distances.
- *Efficiency:* They allow for advanced compression and efficient link sharing using Time Division Multiplexing (TDM).
- *Accuracy & Security:* Easy to implement Error Detection/Correction codes and robust Encryption techniques.
- Easy to process, store and encrypt

Advantages

- *High reliability:* Digital signals maintain their integrity over long distances because they can be regenerated perfectly at intermediate points, resulting in highly reliable communication.
- *Secure and easy to encrypt:* Digital data can be encrypted using mathematical algorithms, making it simple to provide strong security and protect information from unauthorized access.
- *Bandwidth-efficient:* Digital systems use advanced modulation and compression techniques that allow more information to be transmitted within the same bandwidth, improving overall spectrum efficiency.
- *Less affected by noise:* Because digital signals use discrete 0s and 1s, small noise disturbances do not significantly impact the ability of the receiver to correctly interpret the transmitted data.

- *Supports multiplexing (TDM):* Digital communication allows multiple users or data streams to share the same channel using Time Division Multiplexing, improving resource utilization.
- *Allows advanced coding and compression:* Digital signals support various coding schemes and compression algorithms that reduce data size, correct errors, and improve transmission performance.

Limitations

- *Requires analog-to-digital conversion:* Since most real-world signals are analog, they must be converted into digital format before transmission, which adds additional processing steps and complexity.
- *More complex hardware:* Digital communication systems require sophisticated circuitry such as ADCs, DACs, processors, and memory units, making the hardware more complex compared to simple analog systems.
- *Higher processing requirements:* Digital data must be encoded, decoded, compressed, and sometimes encrypted, demanding more computational power and increasing the processing load on communication devices.

Example: Data from a computer, a telegraph signal using Morse code, or digitized voice/video (PCM).

Applications: The Internet, modern digital TV, 4G/5G mobile communication, and data from a computer.

2.4 Why is a digital signal more immune to noise compared to an analog signal?

Noise is any unwanted disturbance added to the original signal during transmission (like static on a radio, or hiss on a phone line). Common examples are the static sound on an AM radio, the hiss on an old telephone line, or the snowy dots on an old TV. Noise is always present in every communication channel.

In an analog signal, which is a smooth and continuous waveform, this noise directly adds to the signal and permanently distorts it. As the signal travels longer distances, it becomes weaker while the noise becomes more dominant, leading to significant loss of quality and corruption of information.

Digital signals overcome this problem effectively. A digital signal contains only two distinct levels – ‘1’ (for example +5 V) and ‘0’ (for example 0 V) – so even if

some noise is added, the receiver can still clearly distinguish between these two levels. The receiver simply regenerates clean, sharp pulses as long as it can correctly identify whether a received value corresponds to a '1' or a '0'. This ability to restore the original signal despite noise is known as noise immunity, and it is one of the biggest advantages of digital communication.

2.5 Comparison: Analog vs Digital Signals

Feature	Analog Signal	Digital Signal
Nature	Continuous in time and amplitude	Discrete (Stepped) in time and amplitude
Values	Infinite	Two (0 and 1)
Representation	Waveform directly proportional to information	Binary bits (0s and 1s)
Noise Effect	High susceptibility; noise is permanent	High noise immunity; can be regenerated
Complexity	Simple hardware, complex signal processing	Complex hardware, simple signal processing
Security	Lower security (difficult to encrypt)	High security (easy to encrypt)
Bandwidth	Less efficient	More efficient
Quality	Degrades with distance	Maintains quality
Applications	FM/AM radio, analog TV	Internet, digital TV, 4G/5G

2.6 Digitization

Digitization is the process of converting information from its natural analog form into a digital (binary) format so that it can be stored, processed, transmitted, and reproduced by digital systems.

Digitization involves taking a continuous analog signal—such as sound, image, temperature, or video—and representing it as a sequence of discrete numerical values. This process allows modern computers, microprocessors, and digital communication systems to handle the information with high accuracy, reliability, and immunity to noise.

Digitization makes analog information compatible with digital technologies. It mainly involves two steps:

- **Sampling:** It is the process of measuring the amplitude (strength) of an analog signal at regular, very short time intervals to convert it into a series of discrete-time values. The sampling rate refers to how many samples are taken per second, expressed in samples per second (Hz). A higher sampling rate captures rapid changes in the signal more accurately and provides a more precise digital representation.
- **Quantization:** Quantization involves approximating each sampled value to the nearest predefined level because digital systems cannot represent infinite precision. Think of measuring height with a ruler marked only in centimeters. You round to the nearest cm. This creates the "steps" of the digital signal. This rounding process introduces a small error known as quantization noise, which increases if fewer quantization levels are used.

After these steps, each quantized (rounded-off) value is given a unique binary code (a group of 1s and 0s), which becomes the final digital representation of the sample. This process is called **encoding**. This stream of bits is your digital signal, ready for transmission.

2.7 Basic Signal Parameters

Amplitude (A)

- The Amplitude is the maximum instantaneous value or intensity of a signal's waveform, measured from the equilibrium (zero) point.
- *Units:* Volts (V) or Amperes (A) for electrical signals; Watts (W) or Decibels (dB) for power/intensity.
- *Significance:* It represents the strength or magnitude of the information carried. A higher amplitude relates to greater power and a longer effective transmission range.

Frequency (f)

- The Frequency is the rate at which the signal oscillates or repeats a complete cycle per unit of time.
- *Formula:* $f = (1 / T)$ where T is the time period of one cycle.
- *Units:* Hertz (Hz) (1 Hz = 1 cycle per second).
- *Significance:* Frequency is critical for Modulation and Multiplexing. In wireless systems, it fundamentally determines the propagation characteristics (e.g., how the signal travels, whether it needs line-of-sight, and its range).

Phase (ϕ)

- The Phase is the angular position of the signal waveform relative to a reference point in time (time zero).
- *Units:* Degrees or Radians.
- *Significance:* Phase is used in advanced communication techniques like Phase Modulation (PM) and Phase Shift Keying (PSK) to encode or shift information onto a carrier wave, particularly in digital systems.

2.8 Frequency Spectrum

The frequency spectrum is the graphical representation of a signal in the frequency domain. It shows the amplitude (or power) contained within each individual frequency that forms the complex signal. According to Fourier analysis, any complex signal—such as voice, music, or a square wave—can be broken down into a combination of simple sine waves of different frequencies and amplitudes. The frequency spectrum simply displays these frequency components, helping us understand the range of frequencies that the signal occupies.

Significance:

- *Helps understand signal composition:* The frequency spectrum makes it easy to visualize how a signal is made by showing all its frequency components clearly.
- *Useful for designing filters:* Engineers can identify which frequencies to allow or suppress, making filter design straightforward.
- *Supports efficient bandwidth allocation:* By observing occupied frequencies, communication systems can allocate bandwidth without overlap or interference.
- *Improves signal analysis and troubleshooting:* Unwanted frequency spikes indicate distortions or faults, helping diagnose problems quickly.
- *Useful for selecting suitable media and bandwidth:* Understanding the spectrum helps in selecting suitable media and bandwidth.

Limitations

- *Requires transformation from time domain:* A mathematical process such as FFT is needed to convert the signal into the frequency domain, adding processing requirements.

- *Does not show time-varying behavior:* The spectrum cannot indicate when in time certain frequencies occur, which limits its use for rapidly changing signals.
- *Not intuitive for general users:* Understanding and interpreting the frequency domain requires some technical background, making it less accessible for beginners.
- *Limited for non-stationary signals:* Signals that change characteristics over time may not be accurately represented by a single spectrum.

Examples

- Voice: 300 Hz to 3.4 kHz
- Music: 20 Hz to 20 kHz
- Video: up to several MHz

2.9 Bandwidth (B)

Definition: Bandwidth is the range of frequencies occupied by a signal or allowed by a communication channel. For a signal, it is the difference between its highest and lowest frequency components. For a channel (like a cable or radio link), it is the range of frequencies it can transmit without excessive loss.

Formula: It is mathematically defined as the difference between the highest frequency f_{max} and the lowest frequency f_{min} contained in the signal or supported by the channel.

$$B = f_{max} - f_{min}$$

Unit: Hertz (Hz).

Significance: Bandwidth is directly proportional to the information-carrying capacity of the channel.

- A higher bandwidth means the channel can carry more information (i.e., a higher data rate) in a given amount of time.
- Determines quality of voice, video, and data.
- The bandwidth of the channel must be at least as wide as the bandwidth of the signal being transmitted.

Typical Bandwidth Requirements:

Service	Required Bandwidth
Voice (telephone)	~3 kHz
AM Radio	10 kHz
FM Radio	200 kHz
Standard video	Several MHz
HD video	Tens of MHz
High-speed data	Depends on Mbps/Gbps

2.10 Baseband vs. Broadband Transmission

These are two fundamental methods of using bandwidth.

Baseband Transmission: The digital or analog signal is transmitted in its original frequency range without shifting it to a higher frequency. It occupies the entire bandwidth of the medium. Only one signal can be sent at a time. Example: Ethernet LANs (using twisted pair cable) send digital pulses directly.

Broadband Transmission: The signal is modulated onto a much higher frequency carrier wave. This allows many independent signals to be transmitted simultaneously over the same medium by using different carrier frequencies (Frequency Division Multiplexing, FDM). Example: Cable TV delivers dozens of channels over one coaxial cable; each channel is on a different carrier frequency.

2.11 Transmission Media

The communication channel or transmission medium is the path through which the signal travels from the transmitter to the receiver. These are broadly categorized into Guided (Wired) and Unguided (Wireless).

2.12 Guided Transmission Media (Wired)

The signal is confined to a solid, physical medium.

A. Twisted Pair Cable

Description: Consists of two insulated copper wires twisted together in a uniform spiral (helical) pattern. The twisting reduces the effects of electromagnetic interference (EMI) and crosstalk.

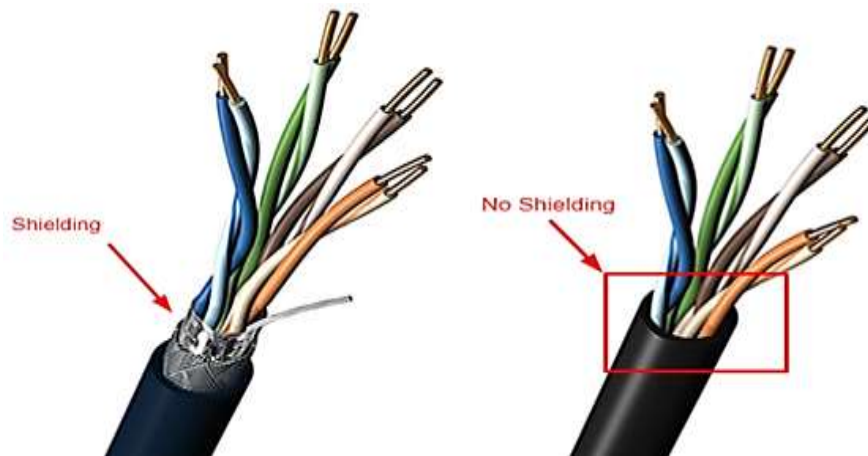
Purpose of Twisting in Twisted Pair Cable

- *Reduces Electromagnetic Interference (EMI):* Twisting ensures that external electromagnetic fields affect both wires equally but oppositely, causing the induced noise to cancel out.

- *Minimizes Crosstalk:* Twisting reduces unwanted signal coupling between adjacent wire pairs, improving signal clarity in multi-pair cables.
- *Improves Signal Quality and Reliability:* By cancelling induced noise, twisting helps maintain better signal integrity over the transmission distance.

Types:

- *Unshielded Twisted Pair (UTP):* Most common for computer networks (LANs), economical, but more susceptible to noise.
- *Shielded Twisted Pair (STP):* Wires are enclosed in a metallic foil or braid, offering better protection from external interference.



Advantages:

- *Low cost:* Twisted pair cables are inexpensive to manufacture and install, making them an economical choice for large-scale communication networks.
- *Easy to install:* Their flexible structure and lightweight design allow twisted pair cables to be easily routed, terminated, and maintained in various indoor environments.

Limitations:

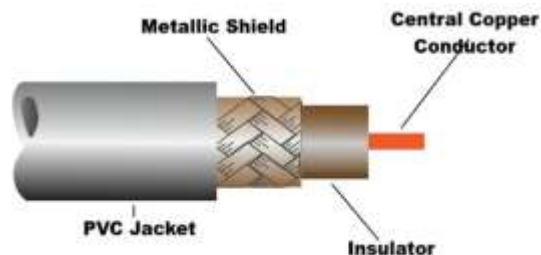
- *Limited bandwidth:* Twisted pair cables support only moderate bandwidth, which restricts their ability to handle high-speed data transmission over long distances.
- *Susceptible to noise:* Because they are not fully shielded, twisted pair cables are vulnerable to electromagnetic interference and crosstalk from nearby electrical sources.

Applications:

- *Telephone networks:* Twisted pair cables are widely used in traditional telephone networks because they can reliably carry voice signals over moderate distances at low cost.
- *Ethernet LAN:* Modern Ethernet LANs commonly use twisted pair cables to connect computers, switches, and routers, providing stable and efficient data communication within buildings.

B. Coaxial Cable

Description: It features a central copper conductor surrounded by an insulator, a metallic shield (braid), and a protective outer plastic jacket. The central conductor carries the signal. The metallic shield provides excellent protection against external EMI and confines the signal's electromagnetic field within the cable.



Advantages:

- *Better noise immunity:* The metallic shield in coaxial cable protects the inner conductor from external electromagnetic interference, ensuring cleaner signal transmission.
- *Higher bandwidth than twisted pair:* Coaxial cables support wider frequency ranges, allowing them to carry more data and higher-quality video signals compared to twisted pair cables.
- *Longer transmission distance:* Due to lower attenuation, coaxial cables can transmit signals over longer distances without requiring frequent amplification.
- *Durable and reliable construction:* The layered structure of coaxial cable makes it physically strong and resistant to damage, ensuring consistent performance in various environments.

Limitations

- *Thicker and less flexible:* Coaxial cables are bulkier and harder to bend than twisted pair cables, making installation in tight spaces more difficult.

- *Higher cost compared to twisted pair:* The multiple layers of insulation and shielding make coaxial cables more expensive to manufacture and install.
- *More difficult to terminate:* Special connectors and tools are required to properly terminate coaxial cables, increasing installation time and complexity.
- *Lower bandwidth than optical fiber:* Although better than twisted pair, coaxial cables cannot match the extremely high bandwidth and performance of optical fiber.

Applications:

- *Cable TV distribution:* Coaxial cables are widely used in cable TV networks because they can carry high-frequency video signals over long distances with minimal interference.
- *Older Ethernet LANs:* Early Ethernet networks used coaxial cables to connect multiple computers in a bus topology due to their good shielding and reliability.
- *CCTV systems:* Coaxial cables are commonly used in CCTV installations because they provide stable transmission of analog video signals with low signal loss.
- *High-frequency data transmission:* Coaxial cables support higher frequencies than twisted pair cables, making them suitable for broadband modems and other high-speed data applications.

C. Optical Fiber Cable (OFC)

Introduction

Optical fiber is a high-speed guided transmission medium that carries information in the form of light pulses through a thin strand of glass or plastic. In the modern digital era, it has become the primary medium for national and international backbone networks because of its ability to handle massive data traffic at high-speed over a long-distance with very low signal degradation. It transmits information in the form of light signals instead of electrical signals.

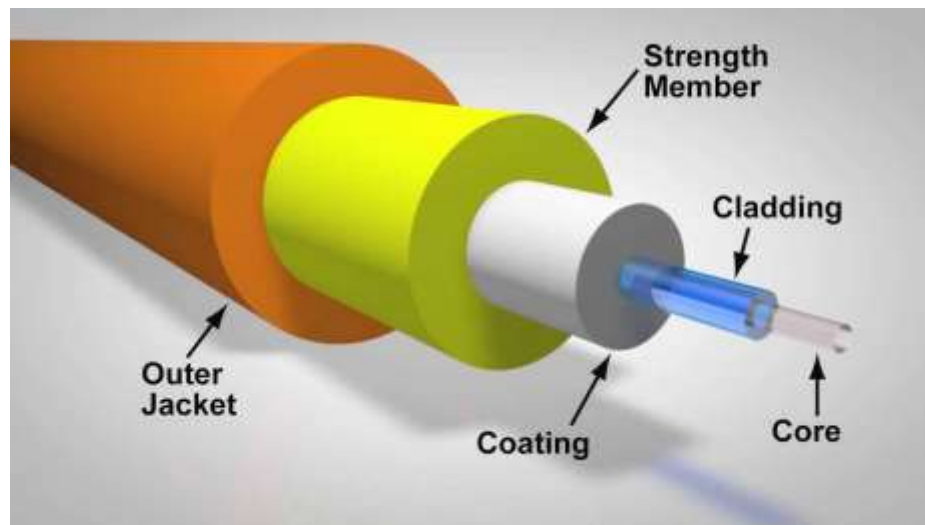
A fiber optic cable consists of very thin strands of glass or plastic, known as optical fibers, which carry information using light waves. Due to low signal loss, high bandwidth, and immunity to electromagnetic interference, fiber optic cables are widely used in telephone networks, internet backbone, cable TV, and data communication systems.

In telecommunication systems, fiber optics has largely replaced copper-based media for long-distance and high-data-rate communication.

Construction of Optical Fiber Cable

An optical fiber cable consists of three main concentric layers designed to protect and guide the light signal:

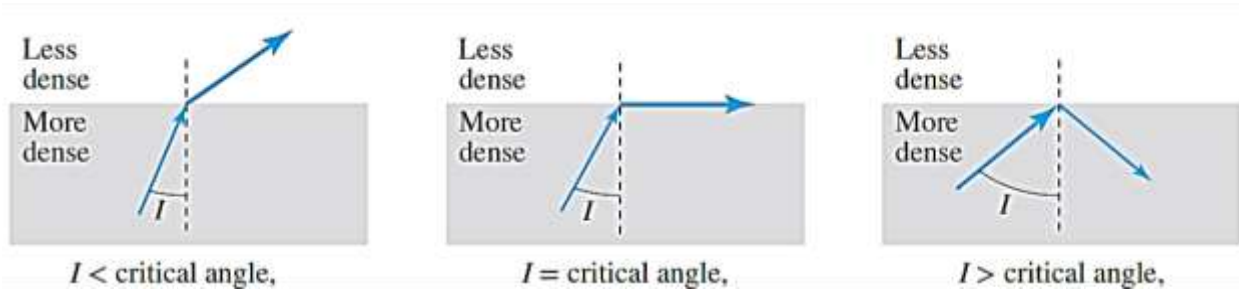
- **Core:** The innermost part made of high-quality glass or plastic with a higher refractive index (n_1). This is the actual region through which light signals travel.
- **Cladding:** The middle layer that surrounds the core. It is made of glass or plastic with a slightly lower refractive index (n_2). The refractive index difference ($n_1 > n_2$) is the fundamental requirement for keeping light trapped inside the core.
- **Buffer Coating/Jacket:** An outer plastic layer that provides mechanical strength and protection from moisture, bending, and external damage.



Fiber optic cable structure

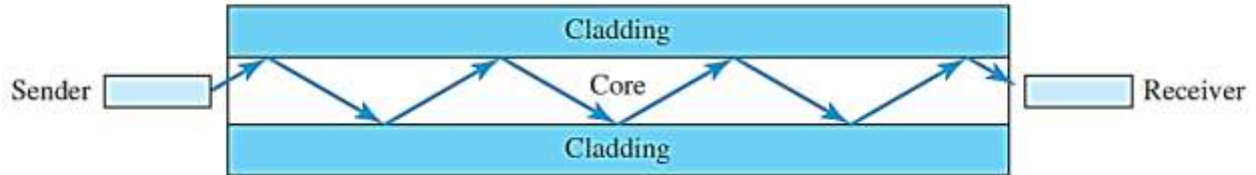
Total Internal Reflection (TIR)

The operation of an optical fiber is based on the principle of Total Internal Reflection (TIR). Light travels in a straight line within a uniform substance. However, when it moves from an optically denser medium (core) to an optically rarer medium (cladding), it changes direction. If the light hits the boundary at a specific angle called the "critical angle," it travels along the interface of the two materials. When light is injected into the core at an angle greater than the critical angle, it does not pass into the cladding. Instead, it reflects completely back into the core. This condition is called total internal reflection (TIR).



Working Principle of Fiber Optic Cable

An electrical signal from the transmitter is converted into a light signal using a light source such as an LED or LASER. The light signal enters the core such that, at the core-cladding interface, the angle of incidence is greater than the critical angle. The core has a higher refractive index than the surrounding cladding. When the light ray strikes the core-cladding interface with an angle greater than the critical angle, total internal reflection occurs. Due to repeated total internal reflections, the light travels along the length of the fiber. At the receiver end, a photodetector converts the received light signal back into an electrical signal.



Types of Fiber Optic Cables

Based on the number of light propagation paths (modes) and the core diameter, fiber optic cables are broadly classified into Single Mode Fiber (SMF) and Multi Mode Fiber (MMF). The choice between these two types depends on factors such as transmission distance, bandwidth requirement, dispersion, and overall system cost.

Single Mode Fiber (SMF)

Single Mode Fiber allows only a single path (mode) of light to travel (propagate) through the fiber core. It has a very small core diameter of approximately 8–10 μm (typically 9 μm) and a standard cladding diameter of 125 μm . Due to the small core size, modal dispersion is almost eliminated, resulting in very low signal distortion and extremely high bandwidth. Single Mode Fiber generally

uses a LASER diode as the light source. Because of its low attenuation and dispersion, SMF is suitable for long-distance and high-speed communication, and is widely used in internet backbone networks, long-haul telecommunication links, and submarine optical cable systems.

Multi-Mode Fiber (MMF)

Multi-Mode Fiber allows multiple modes (rays) of light to propagate simultaneously through the fiber core. It has a comparatively larger core diameter of 50 μm or 62.5 μm , with a cladding diameter of 125 μm . Since multiple light rays travel along different paths, Multi-Mode Fiber suffers from modal dispersion, which limits its bandwidth and transmission distance. MMF typically uses LEDs or LASERS as light sources. Due to easier light coupling, lower system cost, and simpler installation, Multi-Mode Fiber is mainly used for short-distance communication such as Local Area Networks (LANs), data centers, and campus networks.

Comparison between Single Mode Fiber and Multi-Mode Fiber

Parameter	Single Mode Fiber (SMF)	Multi Mode Fiber (MMF)
Core diameter	8–10 μm	50 / 62.5 μm
Number of modes	One	Multiple
Light source	LASER	LED / LASER
Dispersion	Very low	High
Bandwidth	Very high	Moderate
Transmission distance	Very long	Short
Cost	Higher	Lower
Typical use	Long-distance, backbone	LANs, short links

Advantages:

Extremely high bandwidth: Optical fiber supports data rates in the gigabit and terabit range because it carries information using light rather than electrical signals.

- *Very low attenuation:* Signals in optical fiber can travel long distances with minimal loss, reducing the need for repeaters and improving overall efficiency.
- *Immunity to electromagnetic interference:* Because optical fiber transmits light instead of electricity, it is unaffected by electromagnetic noise, ensuring clean and stable communication.

- *Highly secure communication:* It is extremely difficult to tap or intercept optical signals without detection, making fiber an excellent choice for secure data transmission.
- *Lightweight and compact:* Optical fibers are much thinner and lighter than copper cables, allowing easy installation and higher cable density in ducts and conduits.

Limitations

- *Higher initial installation cost:* Optical fiber systems require specialized materials, equipment, and skilled technicians, making the initial deployment more expensive than copper cables.
- *Fragile and requires careful handling:* The glass core of optical fiber is delicate and can break or crack if bent sharply or handled improperly, demanding careful installation practices.
- *Difficult and costly to splice or repair:* Fiber splicing requires precision tools and trained personnel, increasing the complexity and cost of maintenance and troubleshooting.
- *Cannot carry electrical power:* Unlike copper cables, optical fibers cannot transmit electrical power, so separate power cables are needed for active network devices.

Applications:

- *Internet backbone:* Optical fiber is used in national and international communication backbones because it supports extremely high data rates over thousands of kilometers with minimal signal loss.
- *FTTH (Fiber to the Home):* Modern broadband providers use optical fiber directly to homes and offices to deliver high-speed internet, IPTV, and VoIP services with superior reliability.
- *Mobile network backhaul:* 4G and 5G cellular networks use optical fiber to connect base stations to core networks because it provides the high bandwidth and low latency required for mobile data traffic.
- *Long-distance telephone networks:* Optical fiber is used for long-distance telephony as it can carry large amounts of voice and data traffic with excellent clarity and almost no interference.
- *High-speed local networks:* Data centers and enterprise networks use optical fiber for high-speed LAN connections, especially where large volumes of data must be transferred quickly and securely.

2.13 Comparison of Guided Media

Parameter	Twisted Pair	Coaxial	Optical Fiber
Structure	Two insulated copper wires twisted together	Central copper conductor with insulation and metallic shield	Glass or plastic core with cladding and protective jacket
Bandwidth	Low	Moderate	Very high
Data rate	Low to moderate	Moderate to high	Extremely high
Transmission distance	Short	Medium	Very long
Signal attenuation	High	Moderate	Very low
Immunity to EMI	Poor	Better than twisted pair	Excellent (immune)
Crosstalk	High	Low	Nil
Noise susceptibility	High	Moderate	Very low
Security	Low (easy to tap)	Better than twisted pair	Very high (difficult to tap)
Size and weight	Bulky	Thicker and heavier	Thin and lightweight
Installation	Easy	Moderate	Difficult (requires skill)
Cost	Low	Medium	High (initial cost)
Typical applications	Telephone lines, LANs	Cable TV, broadband	Internet backbone, FTTH, long-haul links

2.14 Unguided Media (Wireless)

The signal propagates freely through the atmosphere or space (free space) and is not physically confined. Transmission and reception are facilitated by antennas. A transmitting antenna radiates electromagnetic waves (radio, microwave, etc.) which propagate through the medium (air, vacuum) and are captured by a receiving antenna. An antenna is a transducer that converts electrical signals into electromagnetic waves (transmit mode) and vice-versa (receive mode). A key parameter is gain, which describes its directivity and effectiveness.

Propagation Methods:

- *Ground Wave Propagation:* Signals travel along the earth's surface (e.g., AM radio in MF band).

- *Sky Wave Propagation:* Signals are reflected by the ionosphere back to earth (e.g., HF band for shortwave radio).
- *Line-of-Sight (LOS) Propagation:* Signals travel in a straight line from transmit to receive antenna (e.g., UHF, SHF, EHF bands for TV, mobile, satellite). Requires the antennas to "see" each other.

Advantages

- *Provides Mobility:* It enables communication for moving devices and users, which is the foundation of cellular phone networks and mobile internet, allowing seamless connectivity on the go.
- *Easy Installation Over Difficult Terrain:* It eliminates the need for costly and physically challenging cable trenching or laying across rivers, mountains, or dense urban areas, significantly reducing deployment time and cost.
- *Enables Broadcast (One-to-Many) Communication:* A single transmitter can simultaneously deliver information to an unlimited number of receivers within its coverage area, which is essential for radio, television, and public alert systems.

Disadvantages

- *Susceptible to Interference:* Wireless signals can be disrupted by other transmitters using the same or adjacent frequencies (co-channel/adjacent channel interference) or by non-communication sources like electrical machinery, degrading signal quality.
- *Signal Attenuation Due to Distance and Obstacles:* The signal strength decreases as it travels (path loss) and is further weakened or blocked by physical obstructions like walls, buildings, and foliage, reducing reliable coverage range.
- *Free Space Path Loss:* This is a fundamental, unavoidable weakening of the signal power as it spreads out spherically from the transmitting antenna, governed by the inverse-square law, which limits the effective communication distance.
- *Multipath Propagation (Causing Fading):* The transmitted signal can arrive at the receiver via multiple paths due to reflections from buildings and other objects, causing versions of the signal to combine and sometimes cancel each other out, leading to sudden drops in signal strength (fading).
- *Security Risks (Signals Can Be Intercepted):* Since wireless signals propagate through the open air, they can be received by any unauthorized antenna within range, making eavesdropping and unauthorized access significant concerns that require strong encryption.

Types of Wireless Waves: Radio, Microwave, and Infrared

The broad category of "Wireless Media" is implemented using specific types of electromagnetic waves, each with distinct frequency ranges and applications.

A. Radio Waves

Frequency Range: Generally, 3 kHz to 1 GHz.

Propagation: Can be transmitted in all directions (omnidirectional). Can penetrate walls and travel long distances by diffracting around objects or reflecting off the ionosphere.

Advantages:

- *Long-distance coverage:* Radio waves can travel far, making them suitable for broadcasting and mobile communication over wide geographic areas.
- *Non-line-of-sight communication:* Radio waves can diffract around obstacles and penetrate walls, enabling communication even when transmitter and receiver are not directly aligned.
- *Low-cost implementation:* Radio systems require relatively inexpensive equipment, making wireless communication affordable and widely accessible.
- *Supports mobility:* Radio communication enables users to move freely while staying connected, which is essential for mobile phones and wireless devices.

Limitations

- *Susceptible to interference:* Radio waves easily pick up interference from nearby electronic devices, affecting signal clarity and communication quality.
- *Limited available spectrum:* The radio spectrum is a shared resource, and overcrowding can lead to congestion and reduced performance.
- *Lower bandwidth compared to higher-frequency media:* Radio waves typically support lower bandwidth than microwaves or optical systems, limiting high-speed data capacity.
- *Security risks:* Radio signals radiate in all directions and can be intercepted unless encrypted, making wireless communication vulnerable to unauthorized access.

Applications:

- *AM/FM radio broadcasting:* Radio waves are widely used in AM and FM broadcasting because they can travel long distances and provide reliable audio transmission to large audiences.
- *Mobile communication (2G/3G/4G/5G):* Cellular networks use radio frequencies to connect mobile devices wirelessly to nearby base stations for voice and data services.
- *TV broadcasting:* Radio waves transmit television signals over wide areas, enabling wireless delivery of audio and video content to users.
- *Wi-Fi networks:* Home and office Wi-Fi systems use radio waves to provide short-range wireless internet connectivity between devices and access points.

B. Microwaves

Frequency Range: Generally, 1 GHz to 300 GHz.

Propagation: Highly directional (focused beam). Primarily requires a Line-of-Sight (LOS) path between the transmitter and receiver antennas.

Advantages

- *High bandwidth:* Microwaves support much higher bandwidth compared to radio waves, enabling high-speed communication and data transfer.
- *Highly directional:* Their narrow beam allows focused transmission, reducing interference and improving signal quality between fixed stations.
- *Suitable for long-distance communication:* Microwave links can span tens of kilometers with repeaters, making them ideal for rural or difficult terrains.
- *Less affected by noise:* Microwaves operate at high frequencies where environmental noise is relatively low, improving communication reliability.

Limitations

- *Requires line-of-sight (LOS):* Microwave transmission needs a clear, unobstructed path between antennas, making it sensitive to obstacles such as buildings and hills.
- *Affected by weather:* Heavy rain, fog, and atmospheric conditions can attenuate microwave signals, reducing link performance.
- *Costly antenna installation:* Microwave systems require tall towers and precisely aligned antennas, increasing installation cost and complexity.
- *Limited mobility:* Microwave communication is typically used for fixed-point links and cannot directly support mobile users.

Applications:

- *Point-to-point communication links:* Microwaves are widely used in terrestrial point-to-point links because their highly directional beams allow reliable long-distance communication.
- *Satellite communication:* Microwaves are ideal for satellite links since they can travel through the atmosphere with low absorption and support high data rates.
- *Cellular backhaul networks:* 4G and 5G networks use microwave radio links to connect remote cell towers to the core network where fiber is unavailable.
- *Radar systems:* Microwave frequencies are used in radar systems for navigation, weather monitoring, and speed detection due to their precise reflection properties.

C. Infrared (IR)

Frequency Range: Short-range, very high frequencies 300 GHz – 400 THz.

Propagation: Used for very short-range, line-of-sight communication. Cannot penetrate walls.

Advantages

- *High security within a room:* Since infrared cannot pass through walls, communication remains confined to a single room, reducing the risk of unauthorized access.
- *Free from electromagnetic interference:* Infrared uses light instead of radio waves, making it immune to electrical and RF interference from other devices.
- *Low-cost technology:* Infrared transmitters and receivers are inexpensive to produce, making IR communication affordable for consumer electronics.
- *Simple implementation:* IR systems require minimal hardware and work reliably in short-range, line-of-sight applications.

Limitations

- *Cannot penetrate walls:* Infrared signals are blocked by walls and obstacles, restricting communication to a single room or open space.
- *Requires line-of-sight:* A direct path between the transmitter and receiver is necessary because infrared signals do not diffract around objects.
- *Short communication range:* Infrared systems operate only over a few meters, making them unsuitable for long-distance communication.

- *Affected by sunlight and heat sources:* Strong infrared radiation from sunlight or lamps can interfere with IR receivers, reducing performance in bright environments.

Applications:

- *Remote controls:* Infrared signals are used in TV and appliance remote controls because they provide a simple and reliable short-distance communication method.
- *Short-range device communication:* Infrared is used for short-range data transfer between devices such as laptops, printers, and sensors in controlled indoor environments.
- *Security systems:* Infrared beams are used in alarm and intrusion detection systems because they can detect interruptions in the transmitted IR signal.
- *Indoor wireless links:* Infrared technology provides point-to-point wireless communication within a room where electromagnetic interference needs to be minimized.

2.15 Comparison of Unguided Media

Feature	Radio Waves	Microwaves	Infrared
Frequency	3 kHz – 1 GHz	300 MHz – 300 GHz	300 GHz – 400 THz
Propagation	Omnidirectional	Directional (LOS)	Short Range (LOS)
Wall Penetration	Yes	No	No
Interference	High (RF)	Moderate	Low (Light-based)
Primary Use	Broadcast / Mobile	Satellite / Backhaul	Remotes / Peripherals

2.16 Comparison: Guided Media vs Unguided Media

Feature	Wired (Guided) Media	Wireless (Unguided) Media
Definition	Signals are confined to and travel along a physical solid path (cable).	Signals propagate freely through air, vacuum, or space without a physical conductor.
Examples	Twisted Pair, Coaxial Cable, Optical Fiber.	Radio Waves, Microwaves, Infrared, Satellite links, Cellular networks, Wi-Fi.

Feature	Wired (Guided) Media	Wireless (Unguided) Media
Installation & Mobility	Requires physical cable laying; fixed installation. No mobility.	Easy and quick to deploy over difficult terrain. Supports user/device mobility.
Bandwidth & Data Rate	Very high bandwidth (especially optical fiber); supports extremely high data rates (Gbps to Tbps).	Limited available spectrum; generally lower data rates compared to fiber, though 5G/mmWave offers high speeds.
Attenuation & Range	Signal attenuation is lower and predictable; repeaters extend range to 100s of km (fiber).	Suffers from high free space path loss; signal strength decreases sharply with distance and obstacles.
Interference & Security	High immunity to electromagnetic interference (EMI). Highly secure – physical tapping is difficult and detectable.	Highly susceptible to interference (co-channel, adjacent channel, EMI). Low security – signals can be intercepted; requires encryption.
Propagation Effects	Not applicable. Signal travels within the cable.	Suffers from multipath propagation, fading, shadowing, and atmospheric absorption (rain fade at high frequencies).
Cost Factors	High initial installation and material cost. Low maintenance cost.	Lower initial deployment cost (no trenches/cables). Higher operational cost (spectrum licensing, cell sites).
Topology & Broadcast	Typically, point-to-point or point-to-multipoint (e.g., star in Ethernet). Not suited for broadcast.	Naturally supports broadcast (one-to-many) and multicast communication (e.g., radio, TV).
Reliability	Highly reliable once installed; performance is stable and unaffected by weather.	Less reliable; performance can vary due to environmental conditions, interference, and congestion.
Primary Application	Building network backbones, stable LANs, high-reliability long-distance links (e.g., internet backbone via fiber).	Mobile communication, broadcast services, satellite communication, IoT, and scenarios requiring mobility or quick setup.

Review Questions

Topic 1: Analog vs. Digital Signals

1. Define an analog signal and a digital signal with one example of each from telecommunication.
2. Describe the key characteristics of analog and digital signals.
3. State advantages of digital signals over analog signals for long-distance communication.
4. What is meant by 'digitization' of an analog signal? Mention the role of sampling and quantization.
5. Draw a simple sketch representing a continuous analog wave and a discrete digital binary pulse train.
6. Why is a digital signal more immune to noise compared to an analog signal?
7. Compare analog signals and digital signals.

Topic 2: Bandwidth and Frequency Spectrum

8. Define the term 'Bandwidth' in the context of communication signals. What is its unit?
9. What is the significance of the frequency spectrum in telecommunication?
10. State the relationship between bandwidth and the information-carrying capacity of a channel.
11. Explain the concept of baseband and broadband transmission.

Topic 3: Transmission Media (Wired & Wireless)

12. List the two main categories of transmission media and give one example of each.
13. State the purpose of twisting in twisted pair cable.
14. State two advantages of optical fiber over copper cable.
15. What is the role of an antenna in wireless transmission? State one key parameter of an antenna.
16. Compare Twisted Pair cable and Coaxial cable.
17. State the basic principle of light propagation through an optical fiber. Why is cladding necessary in a fiber optic cable?

18. Compare guided media on the parameter bandwidth, noise immunity, cost and distance.
19. Compare and contrast wired (guided) and wireless (unguided) transmission media.
20. State different propagation methods for electromagnetic waves in telecommunication.
21. List the three main types of electromagnetic waves used in unguided media and state one application for each.
22. Describe the following waves in telecommunication systems with respect to their frequency range, propagation characteristics, advantages, limitations, and applications:
 - (a) Radio waves,
 - (b) Microwaves, and
 - (c) Infrared waves.
23. Describe the following guided transmission media in telecommunication systems with respect to their key characteristics, advantages, limitations, and applications:
 - (a) Twisted pair cable,
 - (b) Coaxial cable
 - (c) Optical fiber cable.